SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 6

Name:
Student ID:

Due on Nov. 10, 2026, 18:00 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(a) For $X \sim \text{Pois}(\lambda)$ and any function g, derive the relationship between $E[Xg(X)]$ and $E[g(X+1)]$. This is called the Stein-Chen identity for the Poisson.

(b) Find the moment $E\left(X^4\right)$ for $X \sim \text{Pois}(\lambda)$ by using the identity from (a) with the fact that X has mean λ and variance λ.

Problem 2

Alice flips a fair coin n times and Bob flips another fair coin $n+1$ times, resulting in independent $X \sim \text{Bin}(n, \frac{1}{2})$ and $Y \sim \text{Bin}(n+1, \frac{1}{2})$.
(a) Let $V = \min(X, Y)$ be the smaller of X and Y, and let $W = \max(X, Y)$ be the larger of X and Y. (If $X = Y$, then $V = W = X = Y$.) Find $E(V) + E(W)$.
(b) Show that $P(X < Y) = P(n - X < n + 1 - Y)$.
(c) Compute $P(X < Y)$. Hint: Use (b) and the fact that X and Y are integer-valued.

Problem 3

Consider two probability distributions $P = (0.75, 0.25)$ and $Q = (0.5, 0.5)$ over a binary random variable $X \in \{0, 1\}$. All logarithms are base 2.

(a) **Basic Calculations (5 points)**

Compute the total variation distance and KL divergence:

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|,$$

$$D_{\mathrm{KL}}(P\|Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}.$$

Is KL divergence symmetric? Justify your answer.

(b) **Connection to Information Theory (5 points)**

i) Compute the entropy $H(P)$ and cross-entropy $H(P, Q)$, which are defined as

$$H(P) = -\sum_x P(x) \log_2 P(x),$$

$$H(P, Q) = -\sum_x P(x) \log_2 Q(x).$$

ii) Show that

$$H(P, Q) - H(P) = D_{\mathrm{KL}}(P\|Q).$$

Briefly explain the significance of this identity in lossless data compression.

(c) **Connection to Machine Learning (5 points)**

Suppose P represents the true label distribution in a binary classification task. Two models predict $Q = (0.5, 0.5)$ and $R = (0.95, 0.05)$, respectively.

Compute $D_{\mathrm{KL}}(P\|Q)$ and $D_{\mathrm{KL}}(P\|R)$. Which model has a smaller expected cross-entropy loss? Explain your answer.